DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 13

✈ Chapter 13: Airborne Weather Radar

AWR — Weather Detection & Ground Mapping

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1. Introduction & Components

Airborne Weather Radar (AWR) provides pilots with weather and navigation information ahead of the aircraft. It requires pilot interpretation — its usefulness depends on operator skill. It can display on a dedicated unit or on the EFIS navigation display (ND).

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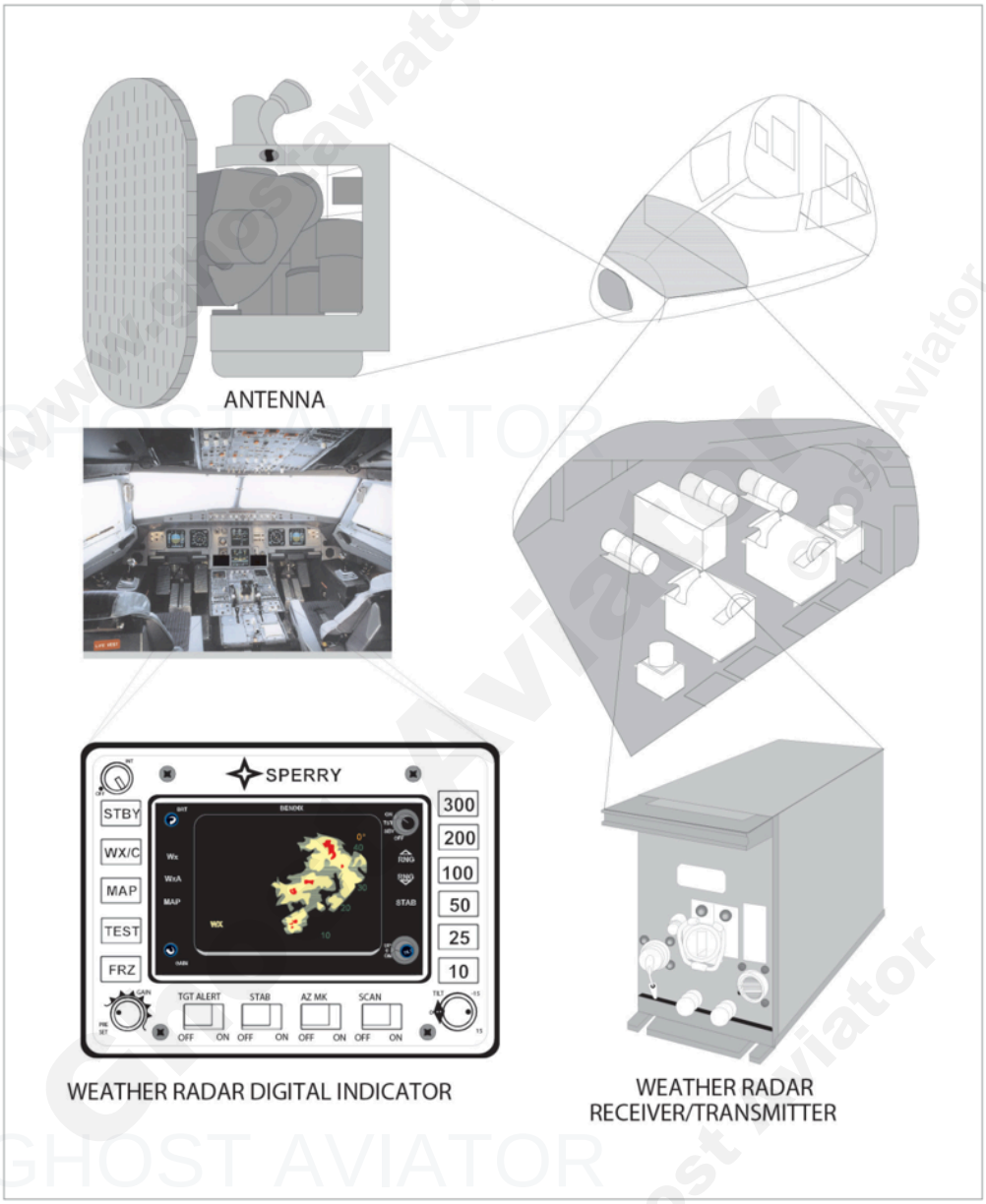


Figure 13.1 AWR Components

13 Airborne Weather Radar

Fig 13.1: AWR components — Tx/Rx, antenna, indicator, control unit

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AWR Components

- **Transmitter/Receiver**

  : generates and detects microwave pulses
- **Antenna**

  : attitude-stabilised in pitch and roll; parabolic or flat plate
- **Indicator**

  : PPI display (monochrome or colour)
- **Control Unit**

  : range, tilt, function, gain controls

2. AWR Functions

Main AWR Functions

- Detect size of water droplets → identify areas of turbulence within clouds
- Determine

  height of cloud tops

  by tilting beam up/down
- Map terrain below aircraft for navigation and high ground avoidance
- Provide position fix (range and bearing) from prominent terrain features

3. Principle of Operation

AWR — Primary Radar Principles

- **Echo principle**

  : determines range of targets
- **Searchlight principle**

  : determines relative bearing of targets
- Range shown on display is

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slant range

— use Pythagoras to find ground range

- Antenna is
  attitude-stabilised
  in pitch and roll — if stabilisation fails display becomes lopsided

AWR Frequency

Why 9375 MHz (3.2 cm)?

- Optimum for detecting
  large water droplets and wet hail
  (~3 cm diameter) associated with severe turbulence
- Typical frequency: **9375 MHz ±30 MHz** (SHF band)
- λ = 300/9375 = **3.2 cm**
- Higher frequency → returns from smaller droplets (clutter); Lower frequency → insufficient returns
- Wavelengths >10 cm produce
  few weather returns

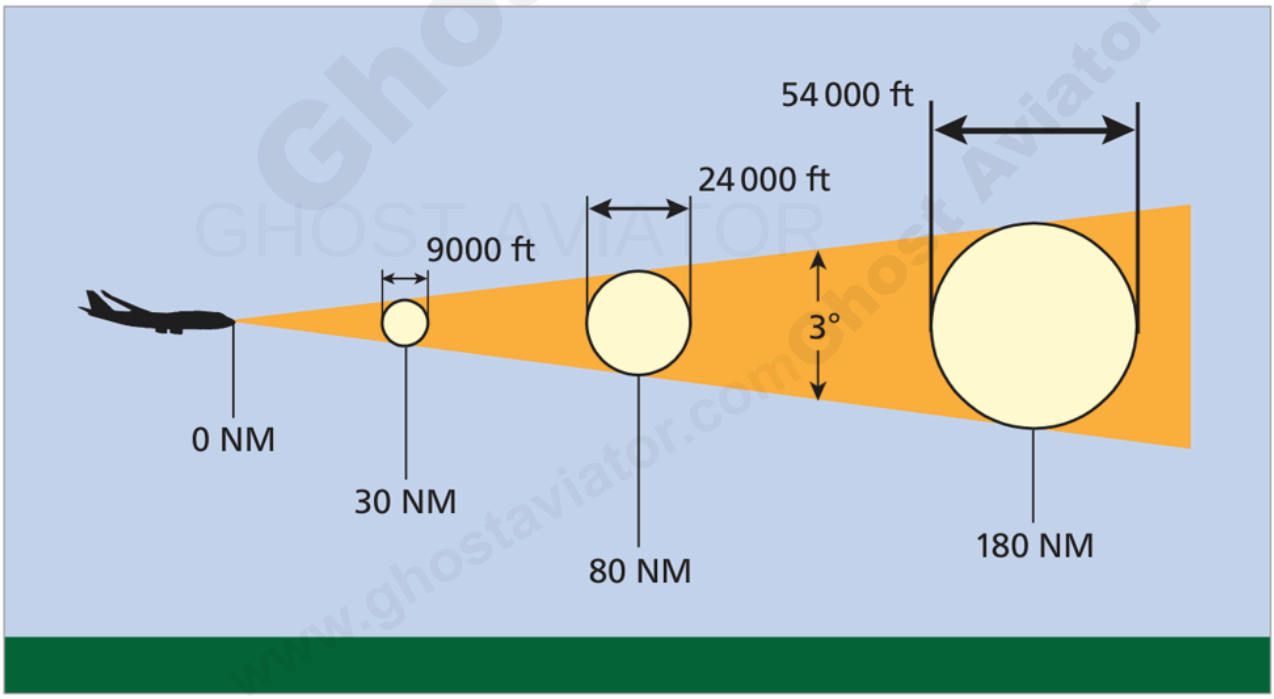


Figure 13.6 *Radar beam coverage at varying ranges*

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Fig 13.6: Radar beam coverage at varying ranges — 3° beam width, range rings at 30/80/180 NM

Radar Beams

Beam Types

Beam	Shape	Used For	Range
Pencil/Conical	3°–5° width	Weather depiction + long range mapping	>60 NM (up to ~150 NM)
Cosecant² / Fan-shaped	85° deep, 3.5° azimuth	Short range ground mapping	15–70 NM

two clouds at say 100 NM might appear as one large return until, at a closer range, they are shown correctly in *Figure 13.5*, as separate entities.

A narrower beam would give better definition but would require a larger antenna which becomes impractical in an aircraft. Therefore, in order to produce the **narrower beams** it is essential to use **shorter wavelengths**.

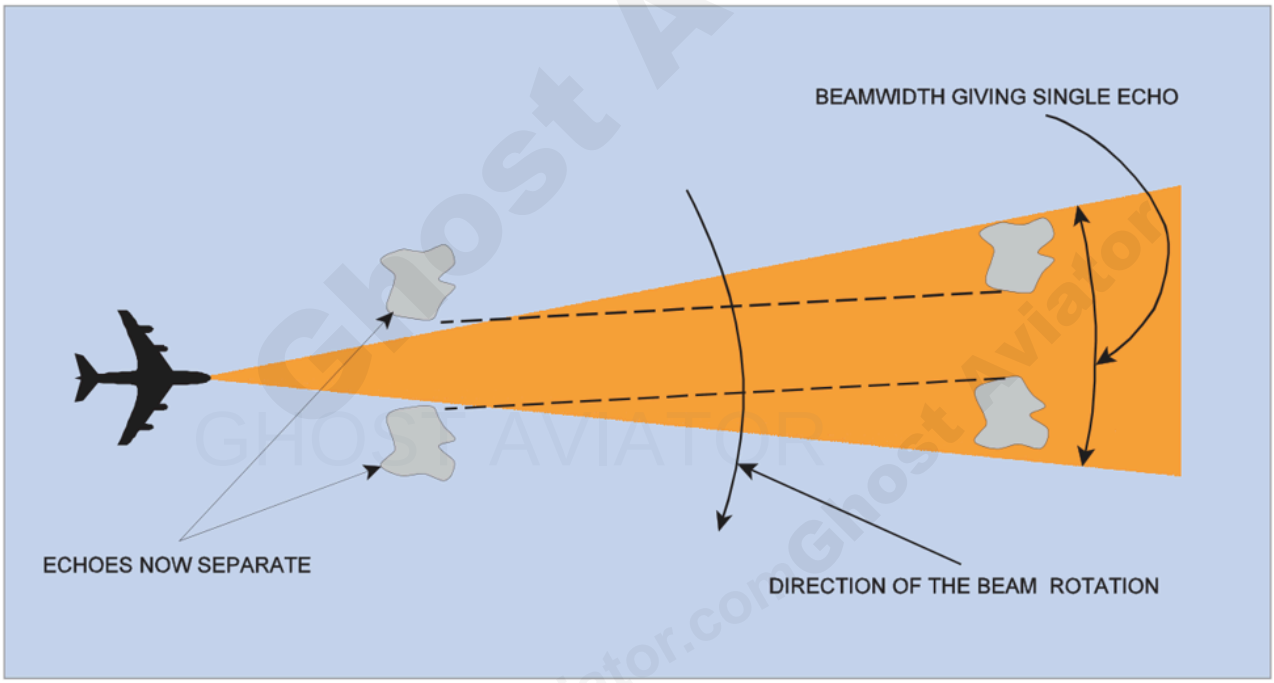


Figure 13.5 Effect of Beamwidth

Fig 13.5: Effect of beamwidth — two clouds at 100 NM appear as one; separate correctly at closer range

Water/Ice in Radome

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Radar energy absorbed by water/ice in the radome (like a microwave oven) → weaker forward transmission → degraded performance. Radome must be kept dry and ice-free.

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4. Weather Depiction & Colours

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Figure 13.2 Monochrome Cloud Display and Avoidance Courses

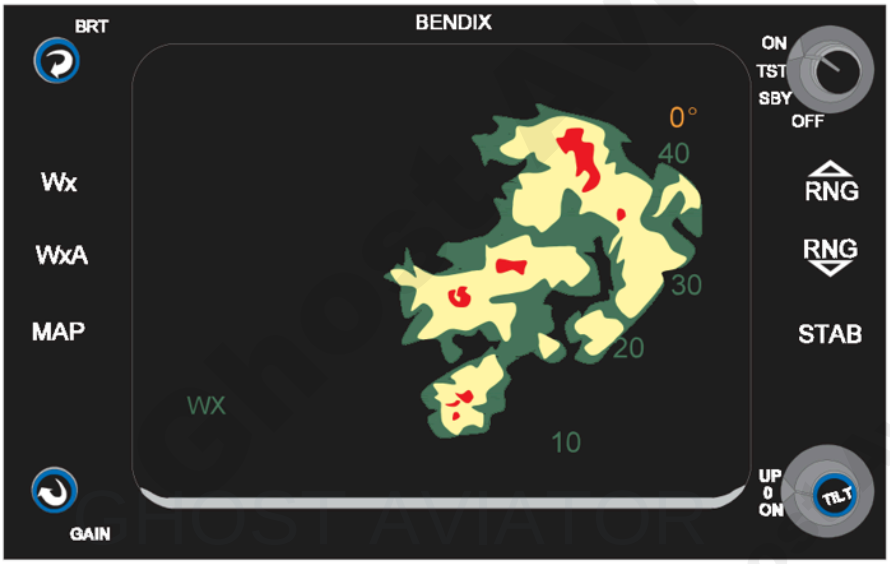


Figure 13.3 Colour Weather Display

Fig 13.2: Monochrome cloud display + avoidance courses | Fig 13.3: Colour weather display

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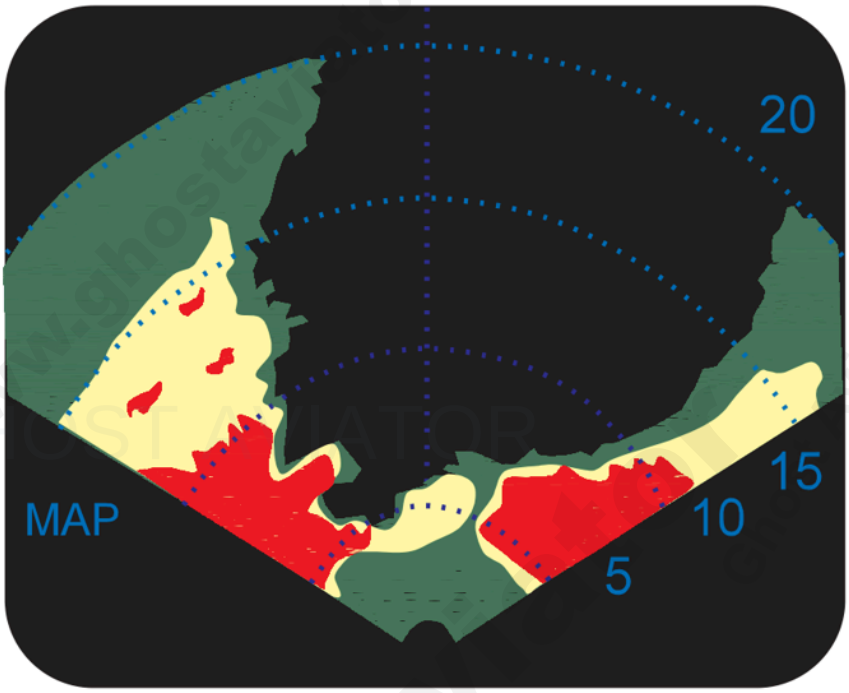


Figure 13.4 Terrain Mapping Display

Fig 13.4: Terrain mapping display — coastlines and built-up areas give bright returns

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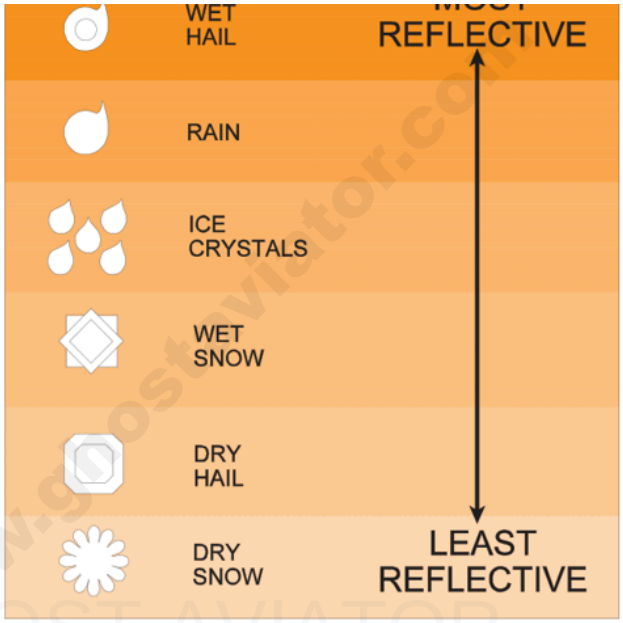


Figure 13.7 Reflective levels

In colour weather radar systems the weather targets are colour-coded according to the intensity of the rainfall as follows:

BLACK	Very light or no returns	Less than 0.7 mm/h.

Fig 13.7: Reflective levels — different precipitation types give different radar returns

Colour Coding — Weather Mode

Colour	Rainfall Rate	Meaning
BLACK	<0.7 mm/h	Very light or no returns
GREEN	0.7–4 mm/h	Light returns
YELLOW	4–12 mm/h	Medium returns
RED	>12 mm/h	Strong returns
MAGENTA	Turbulence intensity	Severe turbulence

On colour systems WITHOUT magenta: RED areas may have a

CYCLIC

function (RED/BLACK alternating) to draw attention.

Colour Coding — Mapping Mode

Colour	Meaning

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GREEN	Light ground returns
YELLOW	Medium ground returns
RED	Heavy ground returns

Turbulence Indicators

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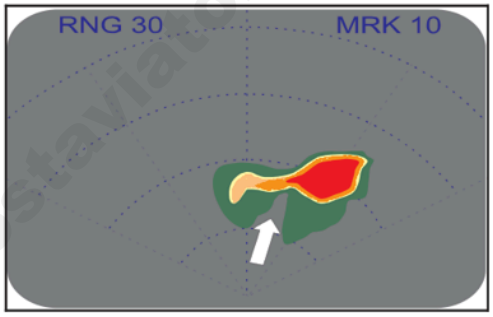


Figure 13.8 U-shape indicating hail activity

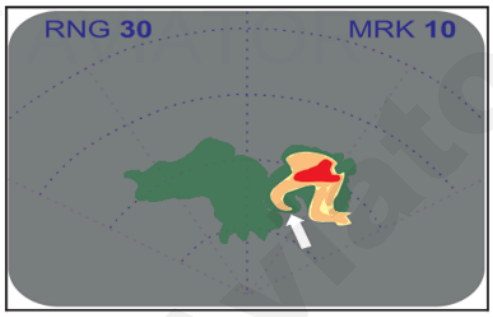


Figure 13.9 Finger indicating hail activity

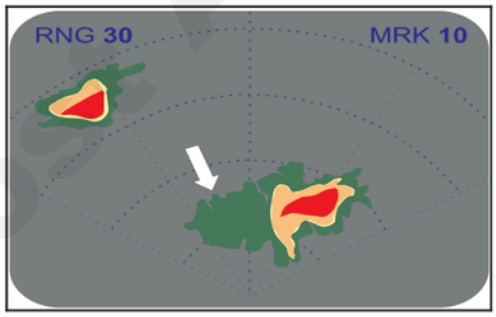


Figure 13.10 Scalloped edge indicating hail activity

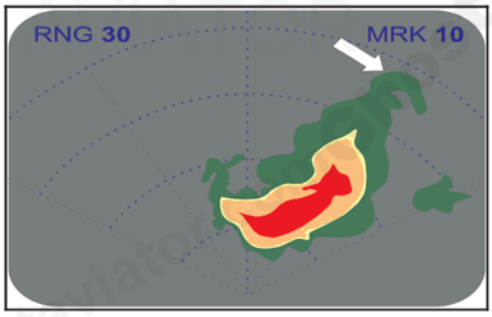


Figure 13.11 Hook indicating hail activity

Fig 13.8–13.11: Turbulence shapes — U-shape, Finger, Scalloped edge, Hook — all indicate hail/severe turbulence

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⚠ Areas to Avoid on AWR Display

- Where colour zones are **closest together** (steepest rainfall gradient)
- **U-shapes** — hail activity
- **Fingers** — hail activity
- **Scalloped edges** — hail activity
- **Hooks** — hail activity

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5. Monochrome Control Unit

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Control Unit

Figure 13.12 illustrates a basic control unit for a monochrome AWR with range scales 20, 50 and 150 NM; its various functions are described below.

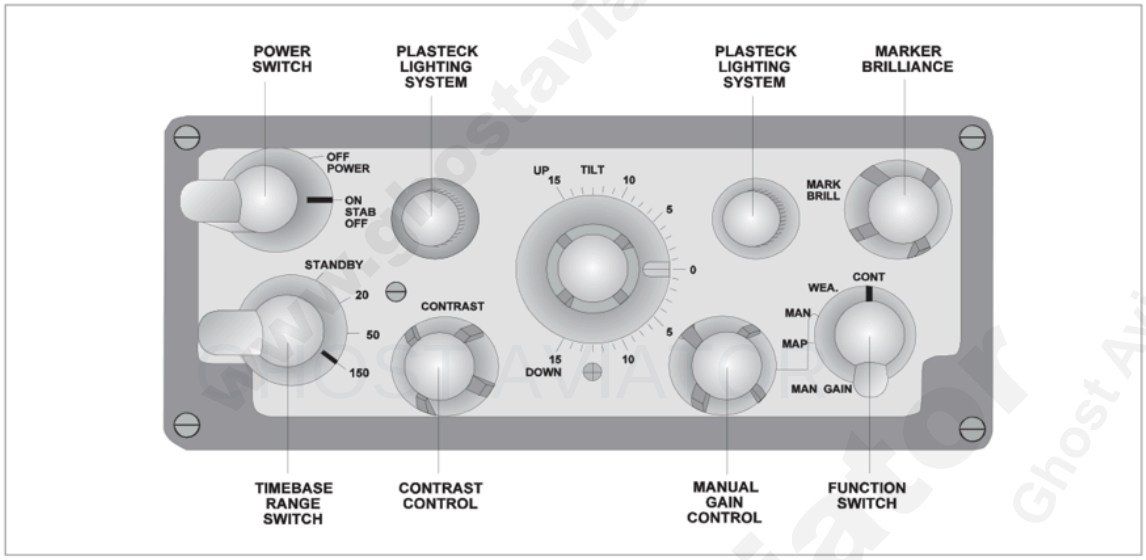


Figure 13.12 Control Unit

Power Switch

In the **ON** position the system is energized and the **aerial** is **automatically stabilized in PITCH and ROLL**. A lopsided or asymmetric display probably indicates that the stabilization has

Fig 13.12: AWR monochrome control unit — power, range, tilt, function switches

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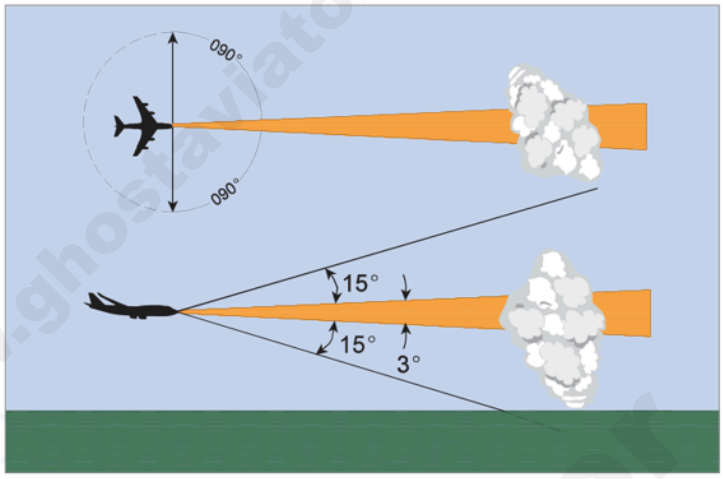


Figure 13.13 *Projected Radar Beam and Tilt Angle*

Function Switch

MAP

In the **MAP** position the radar produces a **mapping beam.** In order to obtain an even presentation of surface features, the transmitted power is progressively reduced as distance decreases so that the power directed to the closest object is minimum. This reduction in power with decreasing range is a function of the cosecant of the depression angle - hence the name **cosecant² beam;** another description is **"fan-shaped"** beam. Its dimensions are 85° deep in the vertical plane and 3.5° in azimuth. Signal amplification is adjustable via the adjacent **MANUAL GAIN** knob.

The minimum (15 NM) and **maximum (60 to 70 NM)** mapping ranges depend upon the aircraft's height and type of terrain. To map **beyond 70 NM the conical pencil beam** should be used by selecting the MANUAL position; this enables the gain to be adjusted, for ground mapping. *See Figure 13.15*.

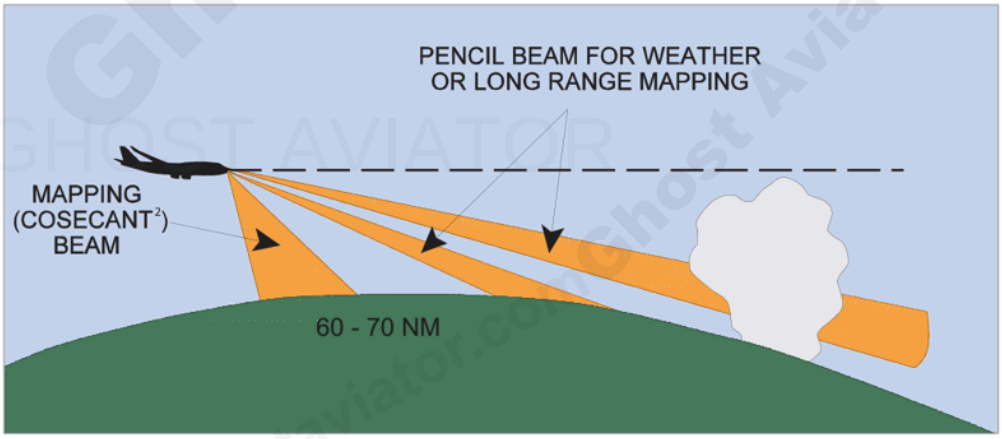


Figure 13.14 *AWR Beam Shapes*

Fig 13.13: Projected radar beam and tilt angle | Fig 13.14: Pencil and fan-shaped beam shapes

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Control Unit Functions

Control	Position	Effect
Power / Stab	ON (Stab On)	System energized; antenna attitude-stabilised in pitch & roll
	STAB OFF	Scanner locked to aircraft axes — display lopsided during manoeuvres
Range	STANDBY	Equipment ready, transmitter OFF. Maintain on ground until clear of personnel
	Range selections	Energizes transmitter; up to **150 NM** practical max
Tilt	±15°	Tilts beam up (+) or down (−). Tilt up for increased range or lower altitude. Sweep ±90° (60° typical)
MAP	Cosecant² beam	Fan-shaped beam; use up to **60–70 NM**. Manual gain operative
MAN	Pencil beam	Manual gain; map >60 NM; weather/navigation 70–150 NM
WEA	Pencil beam	Normal weather observation. AGC/Swept Gain automatic (operative up to ~25 NM) — manual gain INOPERATIVE
CONT	Contour	Highlights dangerous rainfall gradient; Swept Gain also operative; darker colour = dangerous areas

AGC / Swept Gain / STC

Automatic Gain Control decreases receiver gain for nearer echoes → ensures a small nearby cloud does NOT appear brighter than a larger distant cloud. Operates up to ~25 NM. Makes ALL clouds up to 25 NM directly comparable.

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6. Mapping Operation

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Figure 13.16 Hill Shadow

Pre-flight Checks

Electromagnetic radiation presents a serious hazard to personnel, and electronic equipment, therefore great care must be taken before checking the radar on the ground.

The following precautions should be taken:

- Ensure the aircraft is clear of personnel, other aircraft, vehicles and buildings.
- Select conical beam with maximum uptilt, then switch radar on, check you have a picture, then go back to standby.

Fig 13.16: Hill shadow — radar shadow behind mountains may give false impression of water/lakes

Ground Mapping Guide

- MAP (cosecant^2): use up to **60–70 NM**; tilt beam down for best presentation
- MAN (pencil beam): use for mapping beyond 70 NM
- **Bright returns**

  : coastlines, built-up areas, skyscrapers, bridges, power stations
- **Weak returns**

  : calm sea, fine sand, flat terrain
- Ice has jagged edges (reflects);

  snow is a poor reflector

  and masks ground features
- **Hill shadow**

  : mountains create radar shadow → may appear as a false lake

⚠ Pre-flight Radar Check Precautions

- Ensure aircraft is clear of personnel, other aircraft, vehicles and buildings
- Select conical beam with maximum uptilt, switch on briefly, check for picture, return to STANDBY

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- Electromagnetic radiation is a health hazard — never point operating radar at personnel

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7. Weather Operation

✓ Thunderstorm Avoidance Procedure

1. Select maximum range to detect weather in good time; adjust tilt to remove ground returns
2. If storm is extensive, make early track adjustment in consultation with ATC
3. As clouds approach, select lower ranges + CONT; determine best track to avoid turbulence
4. Ensure short-term heading changes steer AWAY from worst areas
5. Constantly switch between short/medium/long ranges to maintain complete storm picture

⚠ AWR Limitations

- **Shadow area**

  : Heavy rain blocks radar waves → area behind heavy rain may conceal severe weather
- **Too high tilt**

  : A thunderstorm may not be detected if tilt is set too high
- **AWR does NOT detect Clear Air Turbulence (CAT)**
- **Height ring**

  : In older AWR with dish antenna, vertical overspill gives a 'height ring' on display — indicates serviceability and roughly the aircraft's height

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8. Colour AWR Controls

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severe turbulence. The Swept Gain facility (or automatic gain control) is also in operation in the CONT position and ensures that a display's intensity does not vary as range decreases.

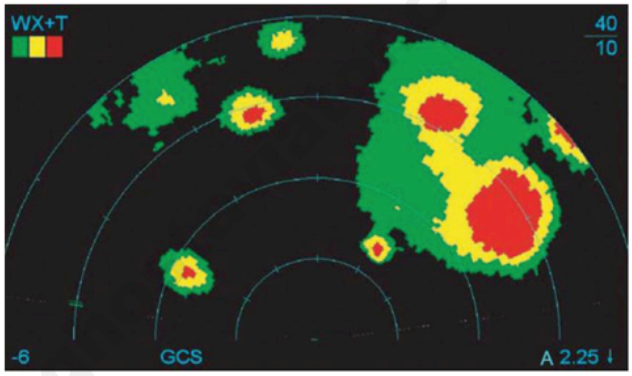


Figure 13.15 Typical cloud display with contour on

Fig 13.15: Cloud display with CONT (contour) on — darker areas indicate dangerous turbulence

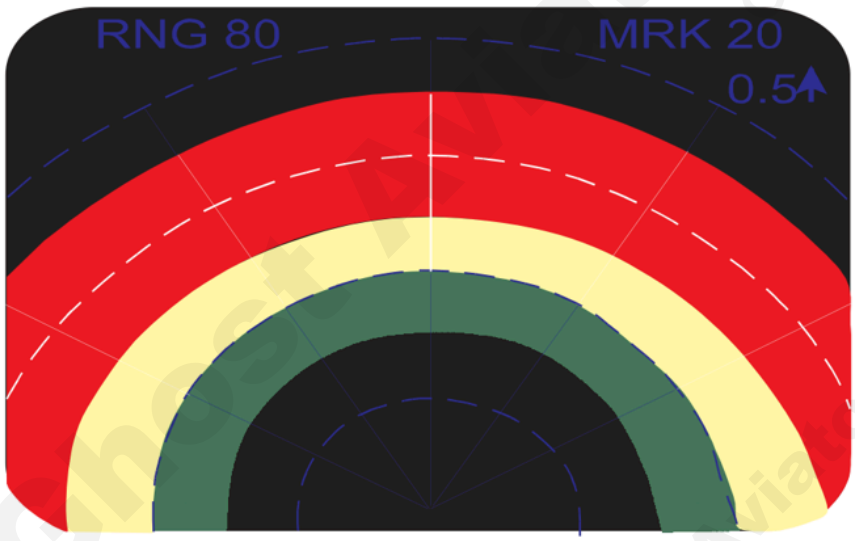


Figure 13.17 Typical Radar System Test Pattern for PPI displays

Hold

This allows the display to be frozen so that storm movements can be assessed. When a storm is located, at say 100 NM, HOLD is selected and a constant heading maintained. HOLD and WX then appear alternately on the screen. After two or three minutes deselect the HOLD facility; this brings back the current display and the storm position is seen to move from its held position to its actual position, thereby indicating its movement relative to the aircraft.

Fig 13.17: Radar test pattern — used for pre-flight serviceability check

Additional Colour AWR Features

Function	Description

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Wx, Wx+T, Wx(var)	Weather / Weather + Turbulence / Weather with variable gain
WxA	Weather Alert — flashing display of turbulence areas
TEST	Displays colour pattern for pre-flight serviceability check
HOLD	Freezes display to assess storm movement. After 2–3 min, deselect to see storm displacement
TGT ALERT	Alerts pilot of contour-strength storm within 60–160 NM and ±15° of heading. Flashes TGT symbol.
FAULT	Flashes on screen if power or transmitter failure detected

9. AWR Summary

Parameter	Detail
Type	Primary radar
Frequency	9375 MHz (±30 MHz), SHF, λ=3.2 cm
Principle	Echo (range) + searchlight (bearing). Range shown is slant range.
Pencil beam	3°–5°; weather + long range mapping (>60 NM)
Fan beam (MAP)	Cosecant^2; short range mapping up to 60–70 NM
Antenna stabilisation	Pitch and roll; lopsided display = stab failure
Tilt	±15°; UP for longer range/lower altitude
Weather colours	Black → Green → Yellow → Red → Magenta
Turbulence warning shapes	U-shapes, Fingers, Scalloped edges, Hooks
AGC range	Operates up to ~25 NM in WEA/CONT mode
Max practical range	~150 NM
AWR does NOT detect	Clear Air Turbulence (CAT)

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10. Practice Questions

Q1. A frequency used by airborne weather radar is:

(a) 8800 MHz

(b) 9.375 GHz

(c) 93.75 GHz

(d) 1213 MHz

Answer: (b)
AWR frequency = **9375 MHz = 9.375 GHz** (SHF band). This gives the best returns from large water droplets (~3 cm) associated with severe turbulence.

Q2. AWR required to detect targets up to 200 NM. Maximum PRR:

(a) 405 pps

(b) 810 pps

(c) 1500 pps

(d) 750 pps

Answer: (a)
Max PRF = 81,000 / 200 = **405 pps**.

Q3. Weather beam preferred over fan-shaped beam for mapping in excess of ___ NM:

(a) 20–25

(b) 60–70

(c) 100–150

(d) 150–200

Answer: (b)
Fan-shaped (cosecant^2) beam used up to **60–70 NM**. Beyond that, use pencil (weather/MAN) beam.

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Q4. AWR is an example of ___ radar at frequency ___ in the ___ band:

(a) primary, 8800 MHz, SHF

(b) secondary, 9.375 MHz, UHF

(c) secondary, 9375 MHz, SHF

(d) primary, 9375 MHz, SHF

Answer: (d)
AWR is **primary radar** at **9375 MHz** in the **SHF** band.

Q5. Correct colour sequence as returns get stronger:

(a) red, yellow, green

(b) yellow, green, red

(c) green, yellow, red

(d) red, green, yellow

Answer: (c)
Colour sequence in order of increasing return strength: **Black → Green → Yellow → Red → Magenta**.

Q6. A false indication of water may be given by AWR when:

(a) flying over land with Land/Sea switch in Sea position

(b) flying over mountainous terrain

(c) cloud/precipitation between aircraft and cloud target

(d) using mapping beam beyond 50 NM

Answer: (b)
Hill shadows created by **mountainous terrain** can give a false impression of water (lakes) on the AWR display.

Q7. AWR operates at 9375 MHz because:

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(a) 8800 MHz gives best returns from all precipitation

(b) 13300 MHz used

(c) 9375 MHz gives best returns from rainfall associated with Cb

(d) 9.375 GHz best for detecting aircraft

Answer: (c)
9375 MHz gives the best returns from **large water droplets and wet hail** found in convective clouds (Cb) associated with severe turbulence.

Q8. Mapping mode of AWR utilizes:

(a) pencil beam from 70 NM to 150 NM

(b) cosecant^2 beam effective to 150 NM

(c) pencil beam max range 70 NM

(d) cosecant^2 beam effective 50 NM to 70 NM

Answer: (a)
MAP (cosecant^2) is used up to 60–70 NM. For mapping beyond 70 NM, select MAN to use the **pencil/weather beam from 70 NM to 150 NM**.

Q9. AWR uses 9 GHz because the wavelength gives reflections only from:

(a) short wavelength produces higher frequency returns

(b) reflects from cloud water droplets of all sizes

(c) wavelength gives reflections only from the larger water droplets

(d) frequency penetrates clouds for good mapping

Answer: (c)
3.2 cm wavelength is comparable to the size of **large water droplets and wet hail** (~3 cm). This gives reflections from dangerous precipitation while ignoring smaller harmless droplets.

Q10. AWR antenna is stabilised:

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(a) in pitch, roll and yaw

(b) in pitch and roll

(c) in pitch and roll whether stab on or off

(d) in pitch and roll only at 0° tilt

Answer: (b)
AWR antenna is stabilised in **pitch and roll**. STAB OFF locks scanner to aircraft axes — display becomes lopsided during manoeuvres.

Q11. AWR colours: very light/no returns ___, light ___, medium ___, strong ___:

(a) black, yellow, green, magenta

(b) black, green, yellow, magenta

(c) grey, green, yellow, red

(d) black, green, yellow, red

Answer: (d)
Colour sequence: **Black** (very light) → **Green** (light) → **Yellow** (medium) → **Red** (strong) → Magenta (turbulence).

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